

Module 09: Channel Capacity Theorems: Nyquist & Shannon

AKTU BCS603 Numerical Core: Mathematical formulation of Nyquist bit rate, Shannon channel capacity, linear vs dB SNR conversions, and solved exam problems.

Data Rate Limits: Nyquist Bit Rate vs Shannon Capacity Theorem

1. Nyquist Bit Rate (Noiseless Channel)

$$C = 2 \cdot B \cdot \log_2(M) \text{ bps}$$

B = Bandwidth (Hertz) | M = Number of Discrete Signal Levels

- Assumption:** Theoretical channel with ZERO noise.
- Baud Rate (Signal Rate) = $2 \cdot B$ signals/second.
- Bits carried per signal element = $\log_2(M)$ bits.
- Example: $B = 3000$ Hz, Binary signal ($M = 2$):
 $C = 2 \cdot 3000 \cdot \log_2(2) = 6,000 \text{ bps} = 6 \text{ kbps}$
- If $M = 4$ levels (2 bits/level): $C = 12,000 \text{ bps} = 12 \text{ kbps}$
- If $M = 16$ levels (4 bits/level): $C = 24,000 \text{ bps} = 24 \text{ kbps}$

Notice: Nyquist allows infinite speed if $M \rightarrow \infty$ (Impossible in real world!)

2. Shannon Capacity (Noisy Real-World Channel)

$$C = B \cdot \log_2(1 + \text{SNR}) \text{ bps}$$

B = Bandwidth (Hertz) | SNR = Signal Power / Noise Power (Linear)

- Assumption:** Real physical channel with Thermal Noise.
- Sets the ABSOLUTE THEORETICAL UPPER BOUND on data rate.
- Converting dB to Linear SNR:
 $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^{(\text{SNR}_{\text{dB}} / 10)}$
- Real Telephone Line Example: $B = 3000$ Hz, $\text{SNR}_{\text{dB}} = 31.62$ dB:
 $\text{SNR} = 10^{3.162} \approx 1452 \Rightarrow \log_2(1453) \approx 10.5 \text{ bits}$
 $C = 3000 \cdot 10.5 \approx 31,500 \text{ bps} = 31.5 \text{ kbps}$ (V.34 Modem limit!)

Shannon capacity CANNOT be exceeded, no matter how many levels M are used!

Engineering Synthesis: How Engineers Combine Shannon & Nyquist to Find Signal Levels (M)

Step 1 (Upper Bound): Use Shannon's formula to find the maximum possible error-free channel capacity: $C_{\text{max}} = B \cdot \log_2(1 + \text{SNR})$

Step 2 (Signal Levels M): Equate Nyquist rate to Shannon Capacity: $2 \cdot B \cdot \log_2(M) = C_{\text{max}} \Rightarrow \log_2(M) = C_{\text{max}} / (2B) \Rightarrow M = 2^{(C_{\text{max}} / 2B)}$

Practical Lesson: Shannon tells us the fundamental physics ceiling; Nyquist tells us how many signal levels (constellation points) we must engineer!

1. Summary Comparison: Nyquist vs Shannon

Dimension	Nyquist Bit Rate	Shannon Channel Capacity
Formula	$C = 2 \cdot B \cdot \log_2(M) \text{ bps}$	$C = B \cdot \log_2(1 + \text{SNR}) \text{ bps}$
Channel Condition	Ideal Noiseless Channel	Realistic Thermal Noisy Channel
Primary Constraint	Number of discrete signal levels (M)	Thermal noise floor (SNR)
Physical Role	Tells how many signal levels must be encoded	Defines unbreakable speed ceiling set by nature

2. Solved Numerical: Combined Shannon & Nyquist

Question (AKTU Exam): Ek communication channel ki bandwidth $B = 1 \text{ MHz}$ hai aur uska SNR 63 hai. (a) Shannon capacity nikalein. (b) Nyquist formula ke mutabiq is data rate ko achieve karne ke liye kitne signal levels (M) chahiye?

Solution:

(a) Shannon Capacity:
 $C = B \cdot \log_2(1 + 63) = 10^6 \cdot \log_2(64) = 10^6 \cdot 6 = 6 \text{ Mbps}$

(b) Signal Levels (M):
 $C = 2 \cdot B \cdot \log_2(M) \Rightarrow 6 \cdot 10^6 = 2 \cdot 10^6 \cdot \log_2(M) \Rightarrow \log_2(M) = 3 \Rightarrow M = 2^3 = 8 \text{ signal levels}$